

Supply Chain & Inventory Management Dashboard

Project Overview

The objective of this project was to analyse supply chain operations, inventory management, and delivery performance using Microsoft Excel. The dashboard provides insights into revenue generation, profitability, inventory health, warehouse utilization, and order fulfilment efficiency, enabling data-driven decision-making.

Business Problem

Supply chain managers often face challenges in monitoring inventory levels, tracking delivery performance, and identifying profitable products and regions. The lack of centralized reporting can lead to stock shortages, delayed deliveries, and reduced profitability.

This project aims to provide a comprehensive dashboard that helps stakeholders:

- Monitor key business KPIs.
- Track inventory health across warehouses.
- Analyse revenue and profit trends.
- Evaluate delivery performance.
- Identify top-performing products and regions.

Dataset Description

The dataset contains transactional and operational supply chain data with the following fields:

Column Name	Description
Order_ID	Unique identifier for each order
Order_Date	Date when order was placed
Delivery_Date	Date when order was delivered
Product_Name	Product ordered
Category	Product category
Supplier	Supplier providing the product
Warehouse	Warehouse fulfilling the order
Region	Customer region

Column Name	Description
Quantity	Quantity ordered
Unit_Price	Price per unit
Shipping_Cost	Cost of shipping
Inventory_Level	Current stock level
Profit	Profit earned from order
Delivery_Status	On Time, Delayed, Critical Delay
Shipping_Time	Delivery duration in days
Total_Revenue	Revenue generated
Total_Cost	Total order cost
Inventory_Health_Alert	Healthy, Reorder, Low Stock

Tools & Techniques Used

Microsoft Excel

- Data Cleaning
- Pivot Tables
- Pivot Charts
- Slicers
- KPI Cards
- Conditional Formatting
- Dashboard Design

Key Performance Indicators (KPIs)

Total Revenue

Measures total sales generated from all orders.

Total Profit

Measures overall profitability of business operations.

Average Shipping Time

Average number of days required to deliver an order.

Gross Profit Margin

Percentage of revenue retained as profit.

On-Time Delivery Rate

Percentage of orders delivered within expected time.

Total Categories

Number of product categories analysed.

Dashboard Analysis

1. Revenue by Region

Analysed revenue contribution from different regions to identify high-performing markets.

Insight:

- North region generated the highest revenue.
- South and West regions showed moderate performance.
- East region contributed comparatively lower revenue.

2. Inventory Health Analysis

Classified inventory into:

- Healthy
- Reorder Required
- Low Stock

Insight:

- Majority of inventory remained healthy.
- Small percentage of products required replenishment.
- Critical stock shortages were limited.

3. Warehouse Inventory Analysis

Compared inventory levels and revenue contribution across warehouses.

Insight:

- Warehouse B managed the highest inventory volume.
- Warehouse B generated the highest revenue.
- Warehouses A and C showed balanced utilization.

4. Supplier Profitability Analysis

Compared supplier-wise revenue and profit.

Insight:

- TechCorp generated the highest revenue and profit.
- Some suppliers generated high revenue but lower profit margins.

5. Delivery Performance Analysis

Orders were classified as:

- On Time
- Delayed
- Critical Delay

Insight:

- Majority of orders were delivered on time.
- Delayed and critical-delay orders represented a smaller portion of total deliveries.

6. Monthly Revenue & Order Trends

Analysed monthly sales and order volume trends.

Insight:

- Revenue remained relatively stable throughout the year.
- Seasonal fluctuations were observed in specific months.
- Order volume followed similar trends as revenue.

7. Top Products by Revenue

Identified highest revenue-generating products.

Insight:

- Electronics products dominated revenue generation.
- Premium products generated significantly higher revenue than other categories.

Business Recommendations

Inventory Optimization

- Replenish products flagged under Low Stock and Reorder categories.
- Maintain safety stock levels for high-demand products.

Delivery Improvement

- Investigate causes of delayed deliveries.
- Optimize logistics routes and supplier coordination.

Revenue Growth

- Focus marketing efforts on high-performing regions.
- Increase stock availability for top-selling products.

Supplier Management

- Strengthen partnerships with high-profit suppliers.
- Review contracts with low-margin suppliers.

Conclusion

The Supply Chain & Inventory Management Dashboard successfully provides a centralized view of operational performance, inventory health, and profitability. By leveraging Excel's analytical capabilities, the dashboard enables stakeholders to make informed decisions regarding inventory planning, supplier management, and delivery optimization.